Curriculum Vitae for Pedro Gomes Pinto

Master's Student in Fundamental Mathematics
Institut Élie Cartan de Lorraine · UFR MIM
Université de Lorraine, Metz

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Education

Université de Lorraine, Metz 2023–present

Master of Science, Mathématiques Fondamentales et Appliquées (research orientation). Coursework in differential geometry, algebraic topology, analysis, algebra and probability. Expected completion 2027.

Université de Lorraine, Metz 2023

Licence (B.Sc.), Mathematics and Applications.

Lycée Fabert, Metz 2020–2022

Classes Préparatoires aux Grandes Écoles (CPGE), intensive preparatory program in mathematics.

Research Interests

Differential geometry and algebraic topology, with a focus on the following areas.

- **Lorentzian geometry.** The global structure of Lorentzian spacetimes and causality theory, in particular global hyperbolicity and chronology protection. The geometry of spacelike foliations and the initial value problem in general relativity.
- **Spin geometry and orientability.** Orientability of smooth manifolds and its physical consequences, such as chirality and spin structures. Clifford algebras, spinor bundles and the Dirac operator.
- **Spectral geometry.** Eigenvalue estimates for the Laplace–Beltrami operator and Reilly-type inequalities, and the way curvature, volume and boundary conditions shape spectral bounds.
- **Topological methods in mathematical physics.** Fiber bundles and foliations over parameter spaces, superspace and the space of 3-geometries, and category- and sheaf-theoretic formalizations of the relations between physical theories.

Research Projects and Theses

Inégalités de type Reilly pour les opérateurs de Laplace–Beltrami et de Schrödinger Apr.–July 2026

Master's memoir (mémoire de M2), written during the end-of-degree research internship at Université de Lorraine under the supervision of Jean-François Grosjean. It studies Reilly-type inequalities and the bounds they give on the first eigenvalue of the Laplace–Beltrami and Schrödinger operators in different classes of ambient manifolds, with attention to how curvature, volume and boundary conditions govern the estimates.

La cohomologie de De Rham et l'orientabilité des variétés lisses 2025

Research memoir at Université de Lorraine, supervised by Hervé Oyono-Oyono, on de Rham cohomology and the orientability of smooth manifolds.

Application de la théorie des nœuds à l'ADN 2023

Bachelor's thesis at Université de Lorraine, supervised by Camille Laurent-Gengoux. A collaborative project on topological invariants of knots and their applications to the topology of DNA.

Lecture Notes (Notes de lecture)

Personal courses in French, written while learning each subject and intended for a reader coming from finite-dimensional Riemannian geometry.

Algèbres de Clifford, spineurs et opérateur de Dirac 2026

About 33 pages. A self-contained account of the path from quadratic forms to the Dirac operator. It covers Clifford algebras and their classification, the groups Pin and Spin, spinor representations and chirality, spin structures and the spinor bundle, and finally the Dirac operator and the Schrödinger–Lichnerowicz formula $D^2 = \nabla^*\nabla + \frac{\mathrm{Scal}}{4}$ with its first geometric consequences. A section is devoted to spinors in physics, and the appendices cover the algebraic prerequisites, principal and associated bundles, and the construction of the first Stiefel–Whitney class. The main sources are Lawson and Michelsohn's *Spin Geometry* and Friedrich's *Dirac Operators in Riemannian Geometry*, whose sign conventions are followed.

Schools, Workshops, and Seminars Attended

SCALPOTMAS 2026 — Masterclass on Scalar Curvature and the Positive Mass Theorem 2026

Seminar on Lie Algebroids and Atiyah Obstruction Classes May 2026

Lecture series given by Panagiotis Batakidis, UFR MIM, Université de Lorraine, Metz.

Teaching and Outreach

To be completed as teaching duties are undertaken.

Other Employment

Carrefour City, Montigny-lès-Metz 2022–2024

Part-time sales associate, in charge of inventory and restocking, the checkout, and digital content creation for the store.

Technical Skills

- **Typesetting.** LaTeX at an advanced level, including TikZ and long structured documents, and Markdown.
- **Programming and tools.** Python, Git and GitHub, Android Studio.
- **Software.** *ResearchLibrary*, a personal Android application for organising and consulting mathematical references, designed, built and deployed independently with Android Studio. The source code is available at github.com/gppedro3714/ResearchLibrary.

Languages

- Portuguese — native
- French — fluent
- English — fluent
- German — intermediate
- Spanish — intermediate
- Luxembourgish — beginner
- Italian — beginner

Last updated September 2026.